

GENEMIND SAMPLES BATCH – 1 SAMPLES
COMPARISSION

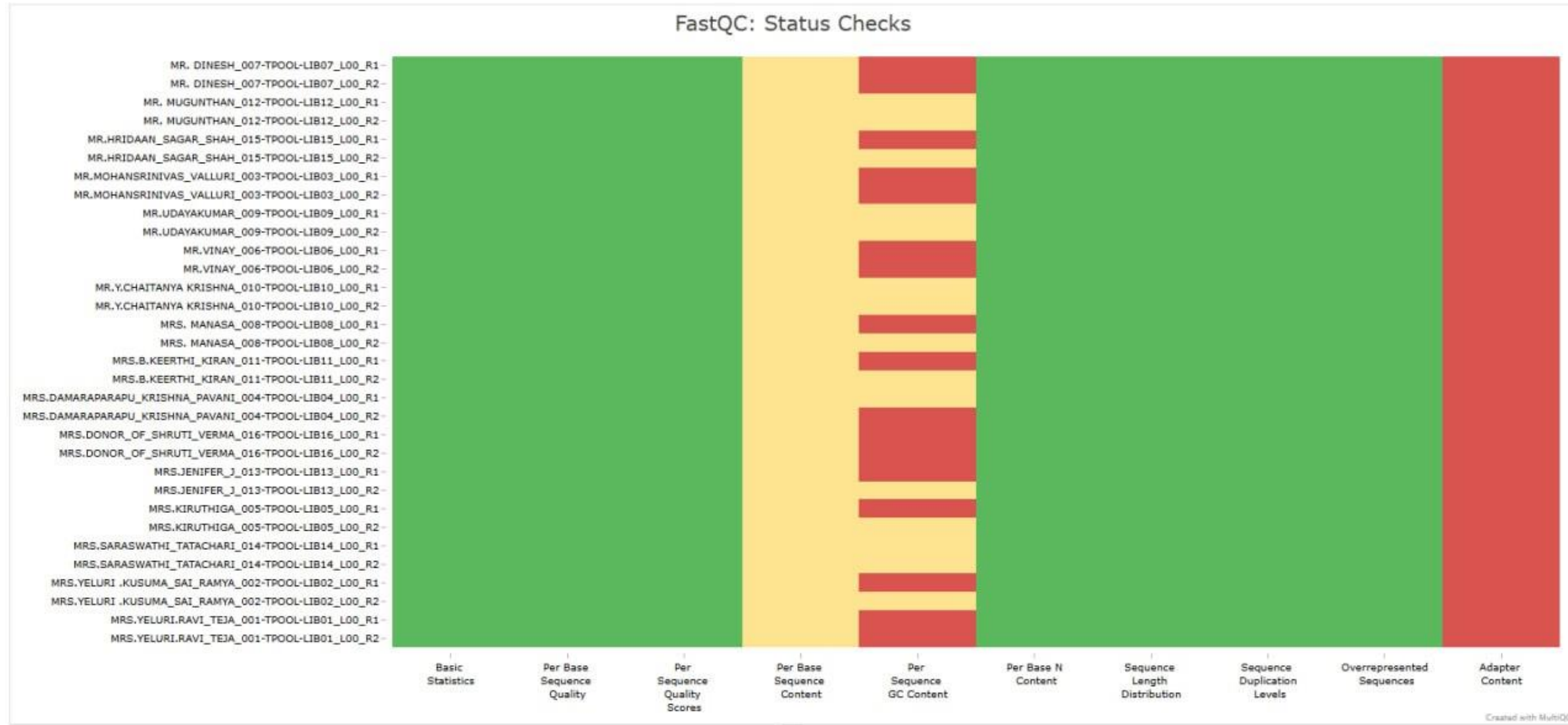
GENEMIND BATCH 15 SAMPLES : (TOTAL SIZE = 84.2 GB)

STEP	INPUT FILE	PROCESS	TOOLS USED	DURATION OF TIME	DATA SIZE
1	FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
2	FASTQ (R1, R2)	FASTP ANALYSIS	FASTP	1 HRS	85 GB
3	TRIMMED FASTQ (R1, R2)	MULTIQC ANALYSIS	MULTIQC, FASTQC	2 HRS	14 MB
4	TRIMMED FASTQ (R1, R2)	HS METRICS ANALYSIS	SENTIEON	4 HRS	80 GB
5	TRIMMED FASTQ (R1, R2)	SECONDARY ANALYSIS	SENTIEON	5 HRS	100 GB
6	VCF, BAM AND BED	SAMPLE SEX AND COVERAGE ANALYSIS	VARSEQ VS CLINICAL COVERAGE AND SAMPLE STASTICS	3 DAYS	10 GB
7	BAM AND VCF	TERTIARY ANALYSIS AND REPORT PREPARATION	VARSEQ	3 DAYS	30 GB
TOTAL				~ 8 DAYS	~ 310 GB

STEP : 1 – MULTIQC ANALYSIS BEFORE TRIMMING

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT BEFORE TRIMMING:



STEP 2 : FASTP (ADAPTER TRIMMING OF FASTQ FILES)

- INPUT FILES : R1, R2 FASTQ FILES
- TOOL : FASTP
- TIME TAKEN: (1 HRS FOR 16 SAMPLES)
- DATA SIZE : 85 GB (FINAL OUTPUT)

FASTQC RESULT:

ADAPTER TRIMMING OF FASTQ FILES

BEFORE AND AFTER FASTP READS COMPARISSION:

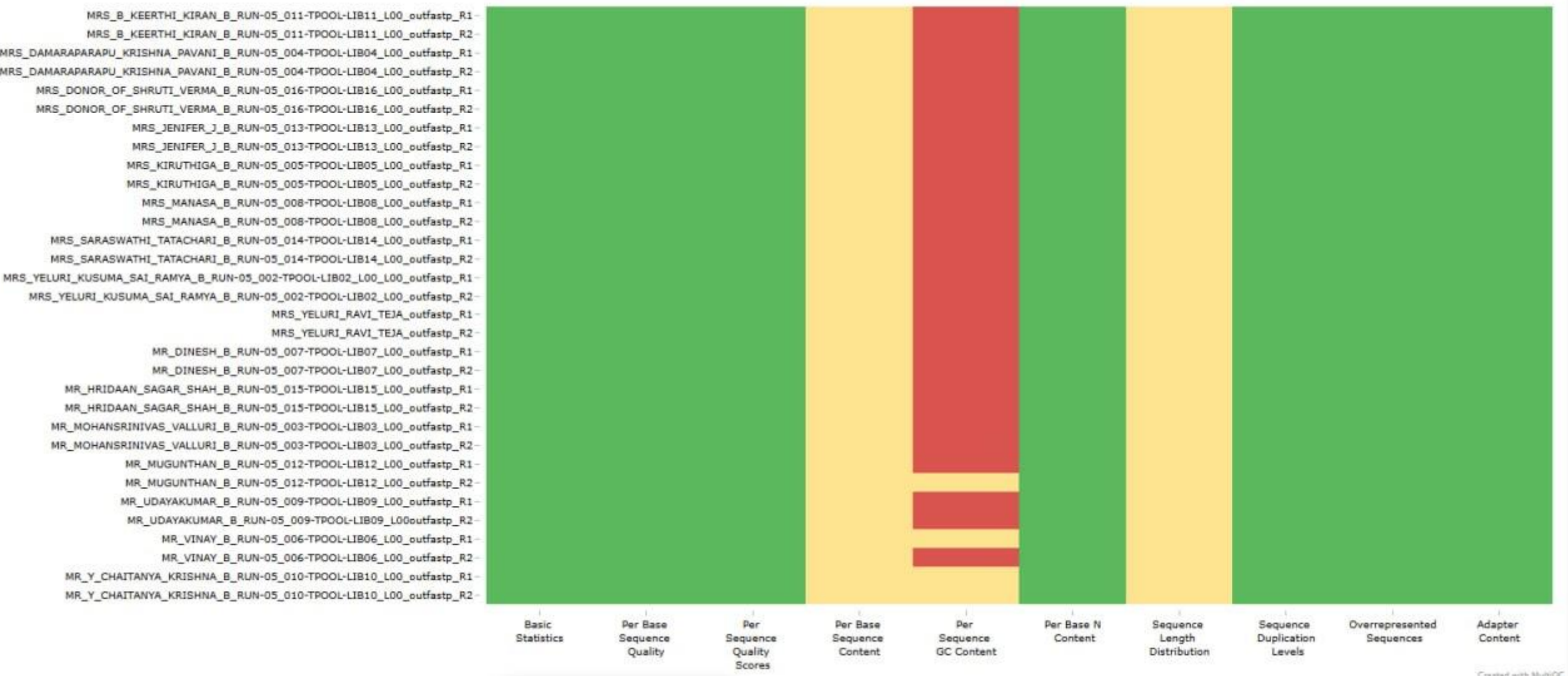
FASTP COMPARISSION RUN 5									
BEFORE FASTP					AFTER FASTP				
Sample_ID	Sample_Name	NUMBER OF READS	SIZE OF READS	Q30	TOTAL DATA GENERATED	NUMBER OF READS	SIZE OF READS	Q30	TOTAL DATA GENERATED
001-TPOOL-LIB01	MRS. YELURI. RAVI TEJA	59.855350 M	150bp	89.93%	8.978302 G	59.729690 M	116bp	91.00%	6.985421 G
002-TPOOL-LIB02	MRS. YELURI . KUSUMA SAI RAMYA	66.708702 M	150bp	90.44%	10.006305 G	66.544408 M	129bp	90.98%	8.633076 G
003-TPOOL-LIB03	MR. MOHANSRINIVAS VALLURI	80.235226 M	150bp	90.55%	12.035284 G	80.039558 M	128bp	91.20%	10.287256 G
004-TPOOL-LIB04	MRS. DAMARAPARAPU KRISHNA PAVANI	55.680764 M	150bp	90.46%	8.352115 G	55.533204 M	133bp	90.83%	7.420122 G
005-TPOOL-LIB05	MRS. KIRUTHIGA	57.566422 M	150bp	90.36%	8.634963 G	57.440188 M	121bp	91.18%	6.972205 G
006-TPOOL-LIB06	MR. VINAY	74.205300 M	150bp	90.13%	11.130795 G	74.013612 M	132bp	90.63%	9.783635 G
007-TPOOL-LIB07	MR. DINESH	84.919976 M	150bp	90.16%	12.737996 G	84.729232 M	122bp	90.94%	10.350919 G
008-TPOOL-LIB08	MRS. MANASA	77.542680 M	150bp	89.93%	11.631402 G	77.373342 M	122bp	90.78%	9.472956 G
009-TPOOL-LIB09	MR. UDAYAKUMAR	66.753158 M	150bp	90.31%	10.012974 G	66.596566 M	127bp	91.05%	8.486233 G
010-TPOOL-LIB10	MR. Y. CHAITANYA KRISHNA	64.527620 M	150bp	90.69%	9.679143 G	64.387466 M	123bp	91.06%	7.975516 G
011-TPOOL-LIB11	MRS. B. KEERTHI KIRAN	67.048874 M	150bp	90.33%	10.057331 G	66.874952 M	133bp	90.83%	8.929926 G
012-TPOOL-LIB12	MR. MUGUNTHAN	69.484364 M	150bp	90.40%	10.422655 G	69.309112 M	132bp	90.87%	9.213237 G
013-TPOOL-LIB13	MRS. JENIFER J	86.070786 M	150bp	89.69%	12.910618 G	85.843516 M	133bp	90.14%	11.465828 G
014-TPOOL-LIB14	MRS. SARASWATHI TATACHARI	71.162210 M	150bp	90.45%	10.674332 G	71.014932 M	118bp	91.21%	8.419290 G
015-TPOOL-LIB15	MR. HRIDAAN SAGAR SHAH	68.300504 M	150bp	90.89%	10.245076 G	68.133014 M	132bp	91.37%	9.038485 G
016-TPOOL-LIB16	MRS. DONOR OF SHRUTI VERMA	68.131042 M	150bp	90.00%	10.219656 G	67.979518 M	123bp	90.79%	8.393501 G

STEP : 3 – MULTIQC ANALYSIS AFTER TRIMMING

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : FASTQC, MULTIQC
- TIME TAKEN: (2 HRS FOR 16 SAMPLES)
- DATA SIZE : 14 MB (FINAL OUTPUT)

FASTQC RESULT AFTER TRIMMING:

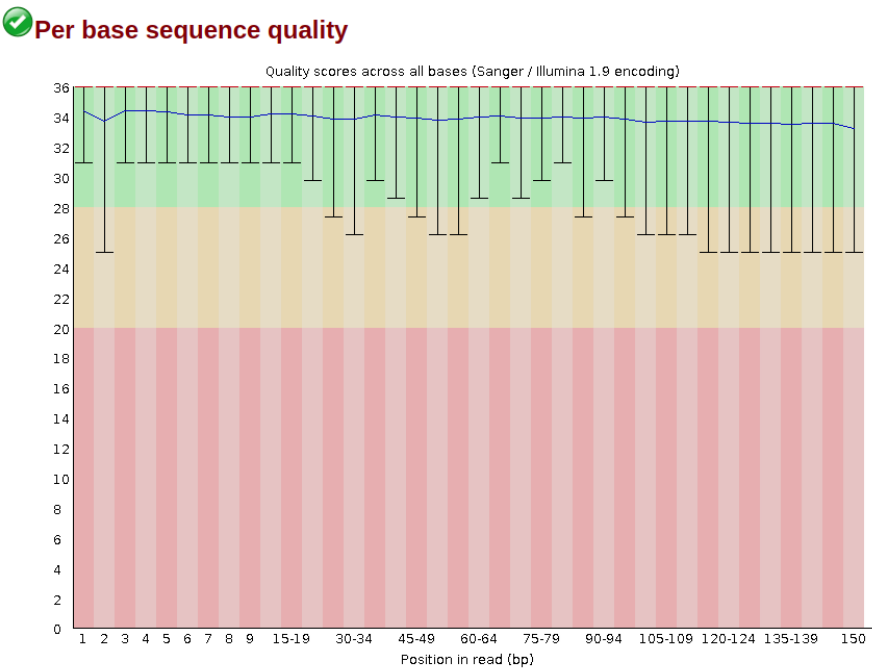
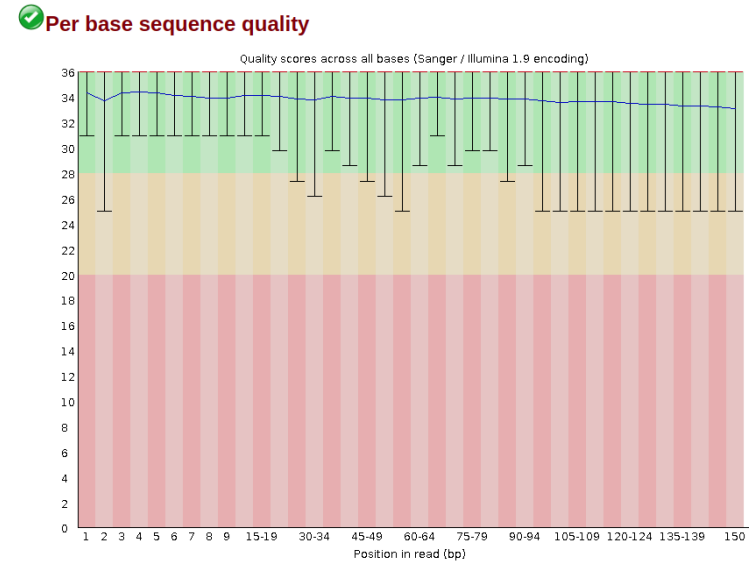
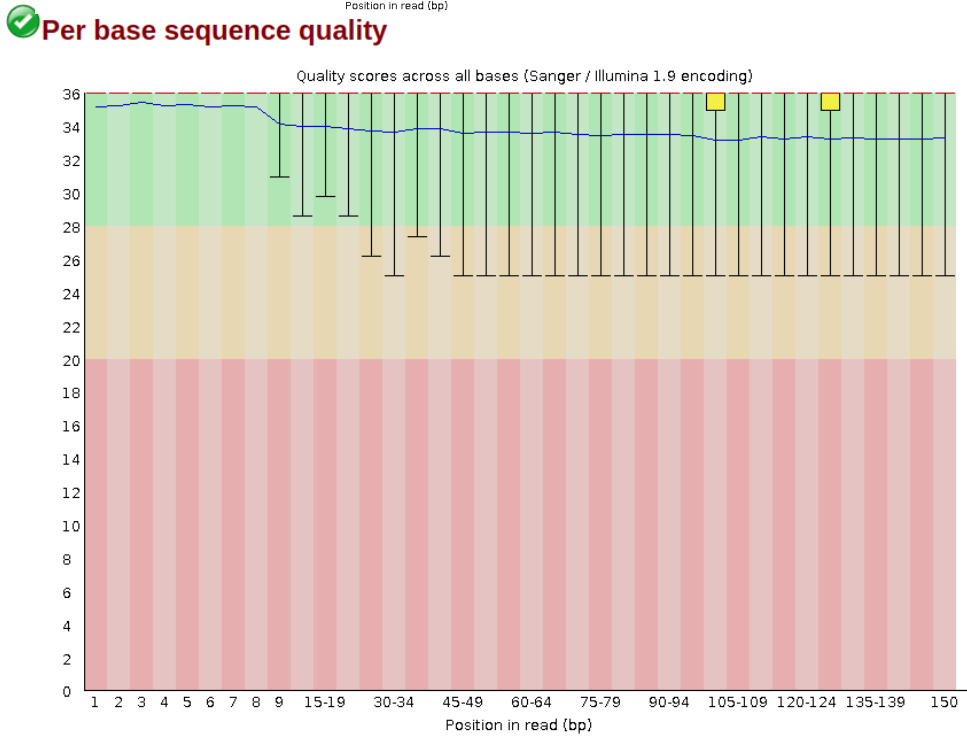
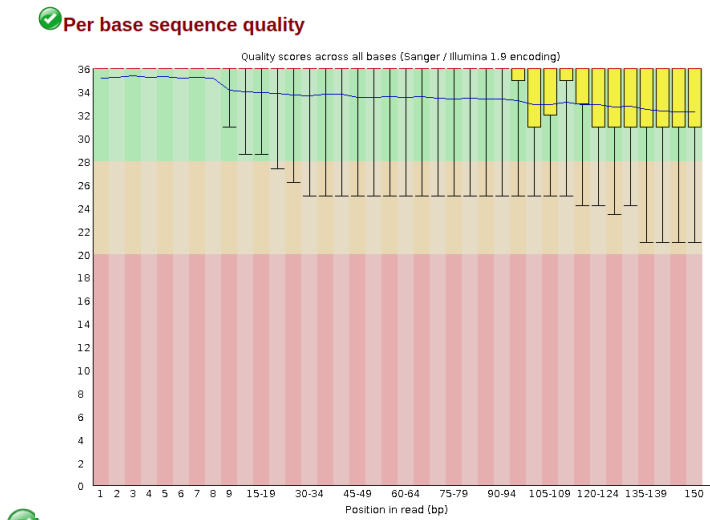
FastQC: Status Checks



FASTQC RESULTS COMPARISSION = BEFORE AND AFTER FASTP(2 SAMPLES)

BEFORE:

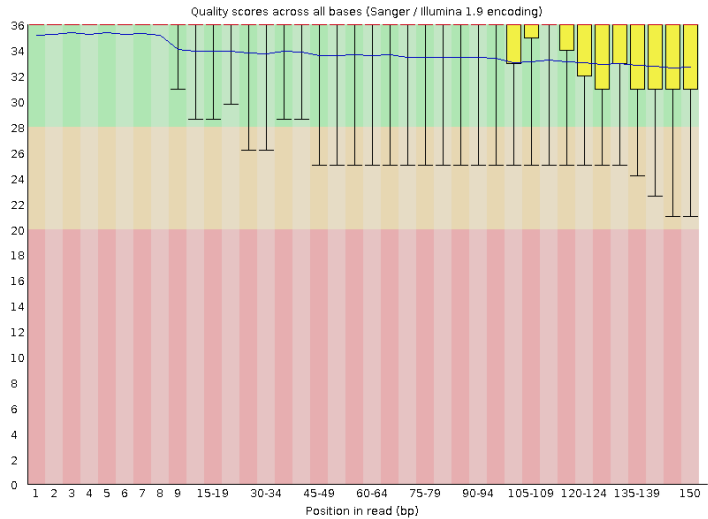
AFTER:



BEFORE:



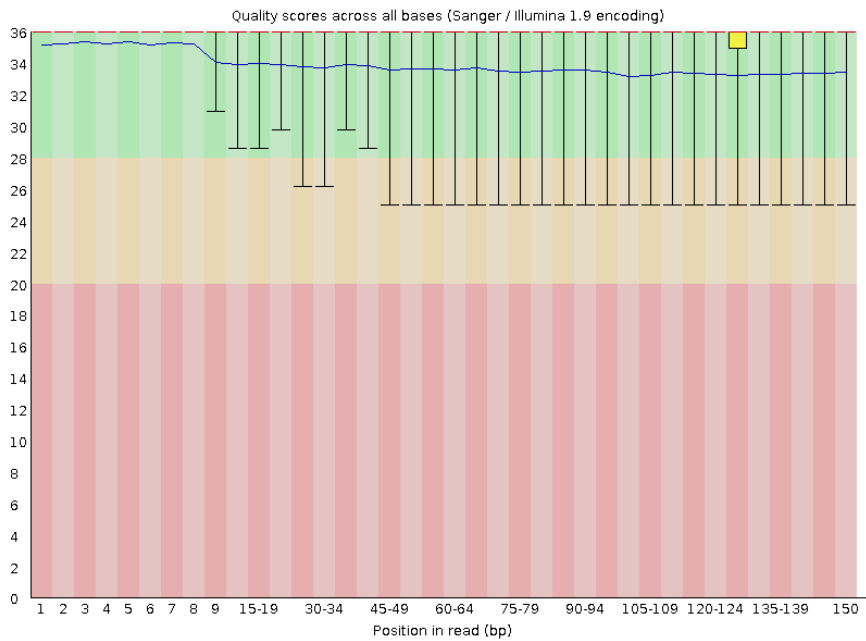
Per base sequence quality



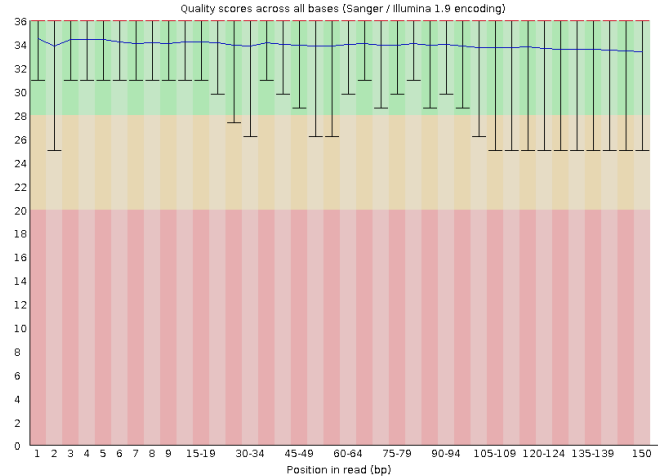
AFTER:



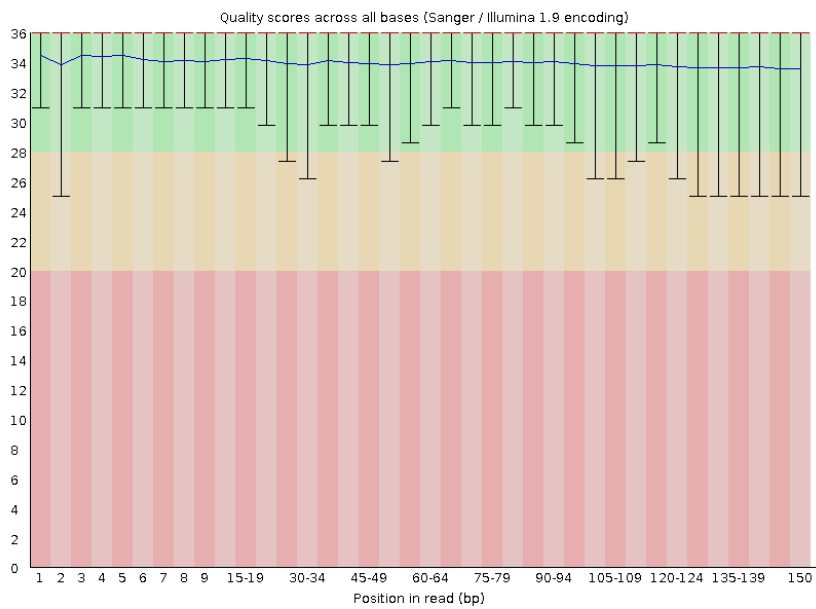
Per base sequence quality



Per base sequence quality



Per base sequence quality



STEP 4 : HS METRICS ANALYSIS :

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (4 HRS FOR 16 SAMPLES)
- DATA SIZE : 80 GB (FINAL OUTPUT)

HS METRICS GENEMIND BATCH – 1:

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000										
BAIT_SET	TWIST = RUN – 5									
	MRS. YELURI. RAVI TEJ	MRS. YELURI . KUSUMA SAI RAMY	IR. MOHANSRINIVAS VALLUR	MRS. DAMARAPARAPU KRISHNA PAVAN	MRS. KIRUTHIGA	MR. VINAY	MR. DINESH	MRS. MANASAMR.	UDAYAKUMAR	
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969
TOTAL_READS	59729690	66544408	80039558	55533204	57440188	74013612	84729232	77373342	66596566	
PF_READS	59729690	66544408	80039558	55533204	57440188	74013612	84729232	77373342	66596566	
PF_UNIQUE_READS	59729690	66544408	80039558	55533204	57440188	74013612	84729232	77373342	66596566	
PCT_PF_READS	1	1	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	59725277	66541659	80036338	55530621	57436803	74009945	84723752	77368169	66593797	
PCT_PF_UQ_READS_ALIGNED	0.999926	0.999959	0.99996	0.999953	0.999941	0.99995	0.999935	0.999933	0.999958	
PF_BASES_ALIGNED	6874119848	8569807871	10185959633	7375505381	6893255864	9724953455	1.0216E+10	9343170814	8414026640	
PF_UQ_BASES_ALIGNED	6874119848	8569807871	10185959633	7375505381	6893255864	9724953455	1.0216E+10	9343170814	8414026640	
ON_BAIT_BASES	4993251550	5924530282	7042222454	5034994115	4936963429	6505092435	7239472152	6667731779	5853190248	
NEAR_BAIT_BASES	1208168385	1796044286	2077854119	1645082062	1298432589	2083505569	1915082115	1757859020	1708810895	
OFF_BAIT_BASES	672699913	849233303	1065883060	695429204	657859846	1136355451	1061845617	917580015	852025497	
ON_TARGET_BASES	3672935616	3994578750	4804938513	3275136333	3523714627	4280355527	5164631653	4757308426	4033914983	
PCT_SELECTED_BASES	0.90214	0.900904	0.895358	0.905711	0.904565	0.883151	0.896065	0.901791	0.898737	
PCT_OFF_BAIT	0.0978598	0.099096	0.104642	0.094289	0.0954353	0.116849	0.103935	0.0982086	0.101263	
ON_BAIT_VS_SELECTED	0.805179	0.767369	0.772167	0.753733	0.791764	0.75741	0.790806	0.791367	0.774027	
MEAN_BAIT_COVERAGE	115.451	136.984	162.827	116.417	114.15	150.407	167.387	154.168	135.334	
MEAN_TARGET_COVERAGE	100.744	109.566	131.793	89.8325	96.8506	117.404	141.859	130.486	110.645	
PCT_USABLE_BASES_ON_BAIT	0.71481	0.68626	0.684558	0.678559	0.708092	0.664895	0.699404	0.70387	0.689728	
PCT_USABLE_BASES_ON_TARGE	0.5258	0.462706	0.467077	0.441386	0.505395	0.437502	0.498954	0.502199	0.475348	
FOLD_ENRICHMENT	54.0356	51.4276	51.4306	50.7833	53.2782	49.7599	52.7136	53.0881	51.7491	
ZERO_CVG_TARGETS_PCT	0.00892788	0.00892788	0.00829043	0.00892436	0.00947729	0.00799459	0.00871305	0.00929063	0.00867078	
PCT_EXC_DUPE	0	0	0	0	0	0	0	0	0	
PCT_EXC_MAPQ	0.041892	0.0373349	0.0388539	0.0370959	0.039968	0.0383743	0.0404667	0.0392416	0.0396446	
PCT_EXC_BASEQ	0.0156928	0.0138025	0.013663	0.0135875	0.0147735	0.0136807	0.0151532	0.0155115	0.0140228	
PCT_EXC_OVERLAP	0.4625	0.442945	0.446466	0.429062	0.456954	0.433973	0.457221	0.458041	0.449132	
PCT_EXC_OFF_TARGET	0.865669	0.794699	0.804222	0.768487	0.83865	0.777687	0.839395	0.841303	0.81167	
FOLD_80_BASE_PENALTY	1.8317	1.68563	1.75724	1.66356	1.78983	1.70151	1.77073	1.78749	1.78459	
PCT_TARGET_BASES_0X%	1.3435	1.3191	1.2129	1.2763	1.3879	1.1658	1.2899	1.3606	1.2517	
PCT_TARGET_BASES_1X	0.986565	0.986809	0.987871	0.987237	0.986121	0.988342	0.987101	0.986394	0.987483	
PCT_TARGET_BASES_2X	0.985902	0.986112	0.987225	0.986499	0.985448	0.987662	0.986513	0.985744	0.986815	
PCT_TARGET_BASES_10X	0.982657	0.983461	0.98493	0.983382	0.982385	0.985113	0.984221	0.983419	0.983972	
PCT_TARGET_BASES_20X	0.973742	0.979378	0.98148	0.976476	0.974608	0.981116	0.981145	0.980014	0.978316	
PCT_TARGET_BASES_30X	0.94955	0.969129	0.973842	0.956058	0.952241	0.971735	0.974458	0.972406	0.963652	
PCT_TARGET_BASES_40X	0.90273	0.945074	0.957712	0.908775	0.905366	0.951467	0.9606	0.955362	0.932958	
PCT_TARGET_BASES_50X	0.833975	0.90043	0.928923	0.830917	0.833177	0.914883	0.936205	0.924207	0.882024	
PCT_TARGET_BASES_100X	0.420145	0.505163	0.624321	0.355529	0.398834	0.558165	0.666312	0.610017	0.497421	
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0	0	
HS_PENALTY_10X	0	0	0	0	0	0	0	0	0	
HS_PENALTY_20X	0	0	0	0	0	0	0	0	0	
HS_PENALTY_30X	0	0	0	0	0	0	0	0	0	
HS_PENALTY_40X	0	0	0	0	0	0	0	0	0	
HS_PENALTY_50X	0	0	0	0	0	0	0	0	0	
HS_PENALTY_100X	0	0	0	0	0	0	0	0	0	
AT_DROPOUT	0.217817	0.0862457	0.113736	0.0841703	0.113344	0.0715294	0.155374	0.140824	0.087366	
GC_DROPOUT	14.9156	13.9224	14.6791	14.0094	14.8528	13.964	14.6029	15.0822	15.0131	

HS METRICS GENEMIND BATCH – 1 :

MIN_MAP=15, MIN_BASE_QUAL_20, COVERAGE_CAP_1000								
BAIT_SET		TWIST = RUN – 5						
MR. Y. CHAITANYA KRISHN/MRS. B. KEERTHI KIRAN/MR. MUGUNTHAN/MRS. JENIFER JMRS. SARASWATHI TATACHA/MR. HRIDAAN SAGAR SHAH/MRS. DONOR OF SHRUTI VERMA								
GENOME_SIZE	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917	3217346917
BAIT_TERRITORY	43249815	43249815	43249815	43249815	43249815	43249815	43249815	43249815
TARGET_TERRITORY	36458262	36458262	36458262	36458262	36458262	36458262	36458262	36458262
BAIT_DESIGN_EFFICIENCY	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969	0.842969
TOTAL_READS	64387466	66874952	69309112	85843516	71014932	68133014	67979518	67979518
PF_READS	64387466	66874952	69309112	85843516	71014932	68133014	67979518	67979518
PF_UNIQUE_READS	64387466	66874952	69309112	85843516	71014932	68133014	67979518	67979518
PCT_PF_READS	1	1	1	1	1	1	1	1
PCT_PF_UQ_READS	1	1	1	1	1	1	1	1
PF_UQ_READS_ALIGNED	64384057	66872277	69306982	85840353	71009282	68130848	67975057	67975057
PCT_PF_UQ_READS_ALIGNED	0.999947	0.99996	0.999969	0.999963	0.99992	0.999968	0.999934	0.999934
PF_BASES_ALIGNED	7869900988	8884394812	9166447952	11391538977	8290730325	8977992322	8278767655	8278767655
PF_UQ_BASES_ALIGNED	7869900988	8884394812	9166447952	11391538977	8290730325	8977992322	8278767655	8278767655
ON_BAIT_BASES	5529744210	6003779919	6186980464	7623720591	5986861012	6103447113	5837724214	5837724214
NEAR_BAIT_BASES	1520449056	1997518849	2033533137	2534810411	1489820301	1988492961	1592209521	1592209521
OFF_BAIT_BASES	819707722	883096044	945934351	1233007975	814049012	886052248	848833920	848833920
ON_TARGET_BASES	3898733589	3900355676	4051365266	4955528154	4378340948	4021976985	4116372403	4116372403
PCT_SELECTED_BASES	0.895843	0.900601	0.896805	0.891761	0.901812	0.901308	0.897469	0.897469
PCT_OFF_BAIT	0.104157	0.0993986	0.103195	0.108239	0.0981879	0.0986916	0.102531	0.102531
ON_BAIT_VS_SELECTED	0.784339	0.750351	0.752627	0.750475	0.800738	0.754263	0.785703	0.785703
MEAN_BAIT_COVERAGE	127.856	138.816	143.052	176.272	138.425	141.121	134.977	134.977
MEAN_TARGET_COVERAGE	106.937	106.981	111.123	135.923	120.092	110.317	112.906	112.906
PCT_USABLE_BASES_ON_BAIT	0.69334	0.672321	0.671532	0.664908	0.711089	0.675273	0.695505	0.695505
PCT_USABLE_BASES_ON_TARGET	0.488838	0.436774	0.439733	0.4322	0.520037	0.444984	0.490424	0.490424
FOLD_ENRICHMENT	52.2696	50.2702	50.2101	49.785	53.718	50.5719	52.4556	52.4556
ZERO_CVG_TARGETS_PCT	0.00849821	0.00876587	0.00811081	0.00868135	0.00937868	0.00806503	0.00928006	0.00928006
PCT_EXC_DUPE	0	0	0	0	0	0	0	0
PCT_EXC_MAPQ	0.0413904	0.037019	0.0371584	0.0366404	0.0405832	0.0384081	0.0394864	0.0394864
PCT_EXC_BASEQ	0.0144676	0.0133548	0.0131895	0.0143944	0.0152854	0.0125925	0.0152484	0.0152484
PCT_EXC_OVERLAP	0.454372	0.426393	0.429571	0.426124	0.461942	0.429375	0.455488	0.455488
PCT_EXC_OFF_TARGET	0.830708	0.767793	0.773429	0.770937	0.860147	0.776355	0.831975	0.831975
FOLD_80_BASE_PENALTY	1.81249	1.67158	1.68369	1.67807	1.79242	1.72371	1.76416	1.76416
PCT_TARGET_BASES_0X%	1.2715	1.2494	1.1739	1.2421	1.3897	1.1756	1.3671	1.3671
PCT_TARGET_BASES_1X	0.987285	0.987506	0.988261	0.987579	0.986103	0.988244	0.986329	0.986329
PCT_TARGET_BASES_2X	0.986586	0.986744	0.987566	0.986875	0.985466	0.987462	0.985694	0.985694
PCT_TARGET_BASES_10X	0.983613	0.983981	0.984677	0.98451	0.982889	0.984447	0.983032	0.983032
PCT_TARGET_BASES_20X	0.976597	0.979631	0.980079	0.981808	0.978492	0.979223	0.978209	0.978209
PCT_TARGET_BASES_30X	0.957733	0.968833	0.968755	0.976872	0.967564	0.966634	0.965666	0.965666
PCT_TARGET_BASES_40X	0.919649	0.943785	0.944309	0.966456	0.94359	0.94038	0.937653	0.937653
PCT_TARGET_BASES_50X	0.860168	0.897154	0.901286	0.945973	0.902205	0.894883	0.889522	0.889522
PCT_TARGET_BASES_100X	0.462759	0.493125	0.519088	0.667113	0.551	0.510117	0.509149	0.509149
HS_LIBRARY_SIZE	0	0	0	0	0	0	0	0
HS_PENALTY_10X	0	0	0	0	0	0	0	0
HS_PENALTY_20X	0	0	0	0	0	0	0	0
HS_PENALTY_30X	0	0	0	0	0	0	0	0
HS_PENALTY_40X	0	0	0	0	0	0	0	0
HS_PENALTY_50X	0	0	0	0	0	0	0	0
HS_PENALTY_100X	0	0	0	0	0	0	0	0
AT_DROPOUT	0.154835	0.0477578	0.0410714	0.0323064	0.113178	0.0267156	0.142145	0.142145
GC_DROPOUT	15.5092	14.0547	13.9881	14.4612	15.0685	14.3127	14.9015	14.9015

STEP 5 : SECONDARY ANALYSIS (BAM AND VCF GENERATION):

- INPUT FILES : TRIMMED R1, R2 FASTQ FILES
- TOOL : SENTIEON
- TIME TAKEN: (5 HRS FOR 16 SAMPLES)
- DATA SIZE : 100 GB (FINAL OUTPUT)

SECONDARY ANALYSIS RESULT:

VCF AND BAM FILES

STEP 7 : VS CLINICAL SAMPLE AND COVERAGE STASTICS:

- INPUT FILES : VCF, BAM, BED FILES
- TOOL : VARSEQ
- TIME TAKEN: (3 DAYS 16 SAMPLES)
- DATA SIZE : 30 GB (FINAL OUTPUT)

SAMPLE STASTISTICS: SAMPLE

SAMPLE STASTICS RUN – 5 TWIST BED								
	MR. YELURI. RAVI TEJA	MRS. YELURI . KUSUMA SAI RAMYA	MR. MOHANSRINIVAS VALLURI	MRS. DAMARAPARAPU KRISHNA PAVANI	MRS. KIRUTHIGA	MR. VINAY	MR. DINESH	MRS. MANASA
Transition Count	118191	137997	167592	116713	113836	201074	161665	139028
Transversion Count	54214	64624	80236	53701	52290	97778	76841	64905
Ti/Tv	2.180082635	2.135383139	2.088738222	2.17338597	2.177012813	2.056433963	2.10388985	2.142022957
Coding Transition Count	20356	19783	20775	20630	20071	20912	20064	20359
Coding Transversion Count	7298	7269	7783	7480	7246	7592	7320	7357
Coding Ti/Tv	2.789257331	2.721557298	2.669279198	2.75802139	2.769942037	2.754478398	2.740983607	2.767296452
Singleton Count	206998	244235	298847	205113	199858	357783	287575	246580
Het Count	69539	74276	86962	81576	71801	91826	77663	78814
Het Ratio	0.335940444	0.304116937	0.290991711	0.39771248	0.359260075	0.256652776	0.270061723	0.319628518
Hom Count	137459	169959	211885	123537	128057	265957	209912	167766
Hom Ratio	0.665374245	0.697287717	0.710417966	0.603641285	0.64209249	0.744495858	0.73126056	0.681573869
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	56965	60126	70544	67185	58681	74942	62327	63863
SNV Hom Count	115263	142315	177041	103082	107274	223628	175951	139935
SNV Het/Hom Ratio	0.494217572	0.422485332	0.398461373	0.651762674	0.547019781	0.335119037	0.354229302	0.456376175
Indel Het Count	11756	13166	15232	13471	12278	15780	14296	14081
Indel Hom Count	22196	27644	34844	20455	20783	42329	33961	27831
Indel Het/Hom Ratio	0.529644981	0.476269715	0.437148433	0.658567587	0.590771303	0.372794066	0.420953447	0.505946606
Var Count	206998	244235	298847	205113	199858	357783	287575	246580
Coding Var Count	28722	28134	29717	29154	28352	29673	28468	28841
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	206998	244235	298847	205113	199858	357783	287575	246580
Call Rate	1	1	1	1	1	1	1	1
SNP Count	172346	202561	247747	170365	166069	298758	238430	203888
MNP Count	0	0	0	0	0	0	0	0
Ins Count	14129	17057	21021	14409	13931	24412	20316	17364
Del Count	20291	24346	29740	20068	19621	34308	28539	25078
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	232	271	339	271	237	305	290	250
Chrom X Het Ratio	0.120721842	0.239130435	0.102075037	0.337550201	0.272141372	0.080868195	0.094221647	0.248690684
Inferred Gender	Male	Male	Male	Female	Male	Male	Male	Male
Multiallelic Site Count	409	492	593	460	421	552	520	435
Multiallelic Het Count	409	492	593	460	421	552	520	435
Coding Multiallelic Site Count	40	32	43	33	35	37	32	28
Coding Multiallelic Het Count	40	32	43	33	35	37	32	28
Q20 Values	97.022319%	97.032954%	97.084455%	96.932053%	97.081208%	96.907468%	96.976515%	96.931299%
Q30 Values	91.00%	90.98%	91.20%	90.83%	91.18%	90.63%	90.94%	90.78%
TOTAL GENES GENERATED	6.985421 G	8.633076 G	10.287256 G	7.420122 G	6.972205 G	9.783635 G	10.350919 G	9.472956 G

SAMPLE STASTISTICS:

SAMPLE STASTICS RUN – 5 TWIST BED								
	MR. UDAYAKUMAR	MR. Y. CHAITANYA KRISHNA	MRS. B. KEERTHI KIRAN	MR. MUGUNTHAN	MRS. JENIFER JM	MRS. SARASWATHI TATACHARI	MR. HRIDAAAN SAGAR SHAH	MRS. DONOR OF SHRUTI VERMA
Transition Count	138294	136746	139049	155537	185354	126254	136818	139082
Transversion Count	65220	64468	65053	74052	88971	58307	63558	65169
Ti/Tv	2.120423183	2.121145374	2.137472522	2.100375412	2.083308044	2.165331778	2.152647975	2.134174224
Coding Transition Count	20612	20306	20846	20970	21163	19750	20831	20706
Coding Transversion Count	7543	7182	7568	7784	7709	7161	7712	7441
Coding Ti/Tv	2.732599761	2.827346143	2.7544926	2.693987667	2.745232845	2.757994693	2.701115145	2.782690499
Singleton Count	246260	242918	246270	276177	329887	223437	242049	246904
Het Count	81499	77116	87251	87902	94577	73311	86585	78988
Het Ratio	0.330946967	0.31745692	0.354290007	0.318281392	0.286695141	0.328105909	0.357716826	0.319913813
Hom Count	164761	165802	159019	188275	235310	150126	155464	167916
Hom Ratio	0.670436049	0.683905723	0.647110529	0.683096292	0.714619776	0.673280204	0.643804586	0.681327956
Hemi Count	0	0	0	0	0	0	0	0
Hemi Ratio	0	0	0	0	0	0	0	0
SNV Het Count	66349	62811	71486	71746	76968	59371	70755	64244
SNV Hom Count	136919	138181	132424	157606	197087	125001	129411	139836
SNV Het/Hom Ratio	0.484585777	0.454555981	0.539826618	0.455223786	0.390528041	0.4749642	0.546746413	0.459423897
Indel Het Count	14134	13337	14699	15042	16395	13020	14686	13844
Indel Hom Count	27842	27621	26595	30669	38223	25125	26053	28080
Indel Het/Hom Ratio	0.507650312	0.482857246	0.552697876	0.490462682	0.428930225	0.518208955	0.563697079	0.493019943
Var Count	246260	242918	246270	276177	329887	223437	242049	246904
Coding Var Count	29277	28536	29522	29890	30062	27977	29655	29254
Ref Count	0	0	0	0	0	0	0	0
Ref Ratio	0	0	0	0	0	0	0	0
Called Genotypes	246260	242918	246270	276177	329887	223437	242049	246904
Call Rate	1	1	1	1	1	1	1	1
SNP Count	203432	201140	204038	229510	274235	184498	200306	204194
MNP Count	0	0	0	0	0	0	0	0
Ins Count	17603	17202	17477	19261	22747	15916	17258	17584
Del Count	24949	24311	24443	27095	32545	22763	24159	24863
DelIns Count	0	0	0	0	0	0	0	0
Complex Count	276	265	312	311	360	260	326	263
Chrom X Het Ratio	0.110522131	0.110590684	0.300690108	0.113956767	0.219132501	0.264590047	0.119427182	0.246390312
Inferred Gender	Male	Male	Female	Male	Male	Male	Male	Male
Multiallelic Site Count	508	484	533	557	607	460	572	450
Multiallelic Het Count	508	484	533	557	607	460	572	450
Coding Multiallelic Site Count	41	31	33	35	35	34	40	37
Coding Multiallelic Het Count	41	31	33	35	35	34	40	37
Q20 Values	97.057829%	97.058368%	96.956184%	97.019872%	96.712015%	97.061096%	97.175106%	96.899070%
Q30 Values	91.05%	91.06%	90.83%	90.87%	90.14%	91.21%	91.37%	90.79%
TOTAL GENES GENERATED	8.486233 G	7.975516 G	8.929926 G	9.213237 G	11.465828 G	8.419290 G	9.038485 G	8.393501 G

COVERAGE STATISTICS :

COVERAGE STASTICS RUN – 5 TWIST BED								
	MR. YELURI. RAVI TEJA	MRS. YELURI . KUSUMA SAI RAMY	MR. MOHANSRINIVAS VALLURI	MRS. DAMARAPARAPU KRISHNA PAVANI	MRS. KIRUTHIG/	MR. VINAY	MR. DINESH	MRS. MANASA
Sample Mean Depth	100.294	122.597	143.298	106.224	100.465	136.141	144.592	133.947
Sample Mean Forward Depth	50.1455	61.2982	71.6486	53.1099	50.2312	68.0687	72.294	66.9726
Sample Mean Reverse Depth	50.1484	61.2986	71.6491	53.1139	50.2333	68.0724	72.2978	66.9743
Sample Mean Filtered Depth	2.77239	3.08097	3.72624	2.58788	2.68774	3.37087	3.83474	3.56199
Sample % 1x	98.6857	98.7111	98.814	98.7504	98.6386	98.8638	98.7419	98.6698
Sample % 5x	98.5015	98.5257	98.643	98.5595	98.4589	98.6852	98.5767	98.5018
Sample % 10x	98.341	98.4099	98.5455	98.4215	98.3163	98.5751	98.4757	98.3975
Sample % 20x	97.6199	98.1382	98.3073	98.0305	97.7594	98.3161	98.2303	98.1361
Sample % 30x	95.5947	97.5688	97.8193	97.0765	96.1485	97.8122	97.7343	97.5925
Sample % 50x	84.9864	93.6169	94.9676	90.3988	86.423	94.824	94.8101	94.0047
Sample % 100x	42.4688	60.4107	69.7733	49.03	43.2393	67.8325	70.0424	64.7058
Sample % 500x	0.0063935	0.0122796	0.0390521	0.0117969	0.00487124	0.0206919	0.0594444	0.0288376
Sample Errors	?	?	?	?	?	?	?	?
Q20 Values	97.022319%	97.032954%	97.084455%	96.932053%	97.081208%	96.907468%	96.976515%	96.931299%
Q30 Values	91.00%	90.98%	91.20%	90.83%	91.18%	90.63%	90.94%	90.78%
TOTAL GENES GENERATED	6.985421 G	8.633076 G	10.287256 G	7.420122 G	6.972205 G	9.783635 G	10.350919 G	9.472956 G

COVERAGE STATISTICS :

COVERAGE STASTICS RUN – 5 TWIST BED								
MR. UDAYAKUMAR MR. Y. CHAITANYA KRISHNA MRS. B. KEERTHI KIRAN MR. MUGUNTHAN MRS. JENIFER J MRS. SARASWATHI TATACHAR MR. HRIDHAAN SAGAR SHAH MRS. DONOR OF SHRUTI VERMA								
Sample Mean Depth	120.695	112.734	126.372	130.557	159.828	120.858	128.618	117.079
Sample Mean Forward Depth	60.3479	56.3658	63.184	65.2777	79.9164	60.4287	64.3103	58.5394
Sample Mean Reverse Depth	60.3467	56.3681	63.1877	65.2791	79.9111	60.4289	64.3078	58.5389
Sample Mean Filtered Depth	3.05171	3.05388	3.07432	3.1656	3.85141	3.31713	3.2146	3.1517
Sample % 1x	98.7742	98.7542	98.7749	98.8521	98.7878	98.6344	98.8541	98.6623
Sample % 5x	98.5966	98.5749	98.5853	98.665	98.6066	98.4682	98.6522	98.489
Sample % 10x	98.4723	98.4407	98.462	98.5399	98.5027	98.3458	98.5167	98.3687
Sample % 20x	98.0914	97.9437	98.1881	98.2481	98.3122	97.9999	98.1796	98.0099
Sample % 30x	97.1705	96.6282	97.6556	97.6636	98.0372	97.1719	97.5063	97.1504
Sample % 50x	91.68	89.1095	94.2582	94.2068	96.6527	91.7913	93.6758	91.4058
Sample % 100x	57.1917	51.0132	63.2452	65.0485	78.7537	57.1771	63.4304	55.0098
Sample % 500x	0.0120821	0.0124881	0.0164626	0.018846	0.0491241	0.0142544	0.020969	0.0134097
Sample Errors	?	?	?	?	?	?	?	?
Q20 Values	97.057829%	97.058368%	96.956184%	97.019872%	96.712015%	97.061096%	97.175106%	96.899070%
Q30 Values	91.05%	91.06%	90.83%	90.87%	90.14%	91.21%	91.37%	90.79%
TOTAL GENES GENERATED	8.486233 G	7.975516 G	8.929926 G	9.213237 G	11.465828 G	8.419290 G	9.038485 G	8.393501 G